International Cybersecurity and Digital Forensics Academy (ICDFA)

Basic Computer Skills for Digital Forensics

Lab 1B: PC System Fundamentals, Disk Sectors and Partition Analysis

SEMESTER 1

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Course Code	CIP-B101
Course Title	Basic Computer Skills for Digital Forensics
Lab Code	Lab 1B
Lab Title	PC System Fundamentals, Disk Sectors and Partition Analysis
Date	7th November 2026
Platform/Tools Used	Windows 11, Kali Linux

Task A - System Identification and Evidence Layer Mapping

A (i)- Capture basic system specification

- Capture a screenshot showing OS Name, Version, System Type, Processor, BIOS Mode, Installed Physical Memory, and Secure Boot State if available.
- Also run the command below in Command Prompt and capture the first page of output.

```
systeminfo
```

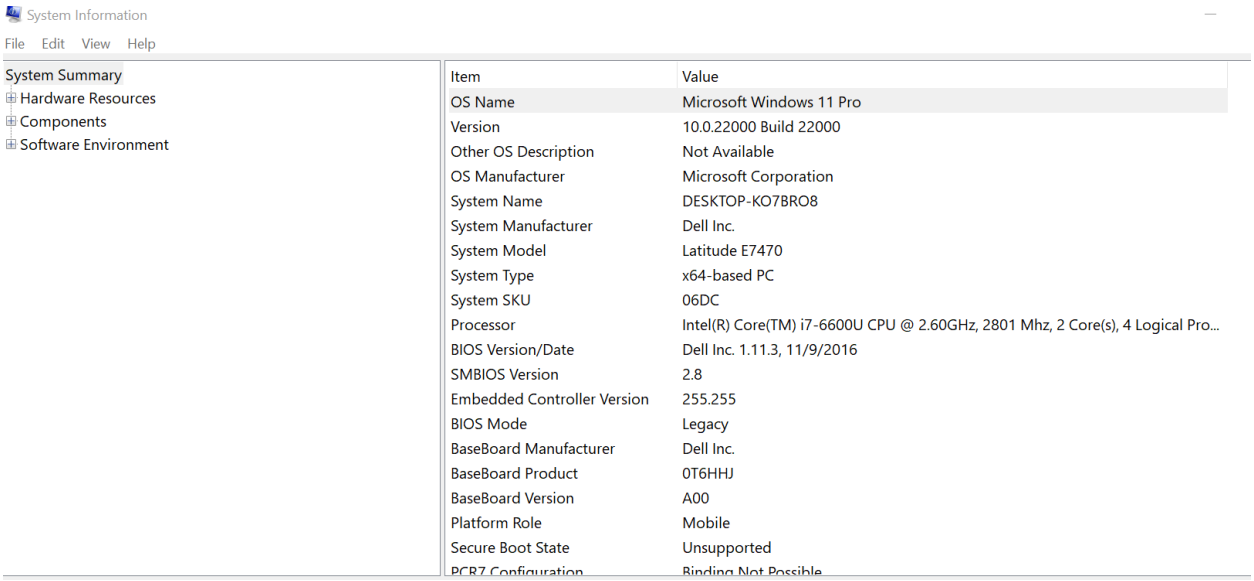


Figure 1- My Windows System Specification (a)

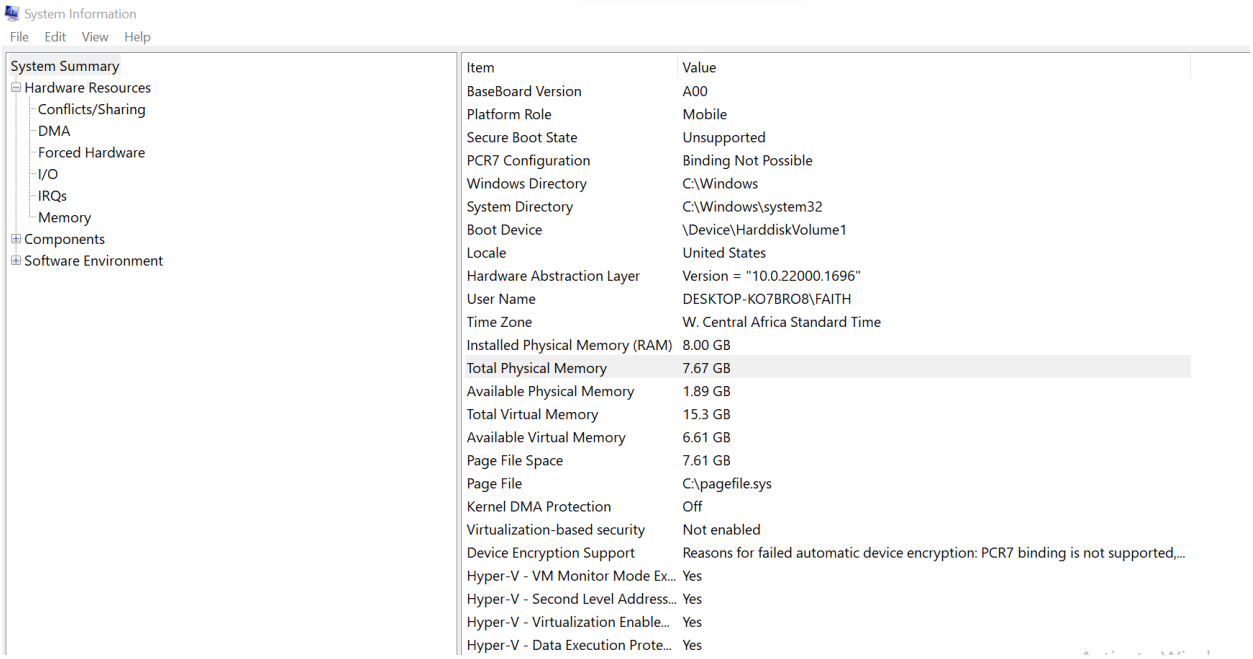


Figure 2- My Windows System Specification (b)

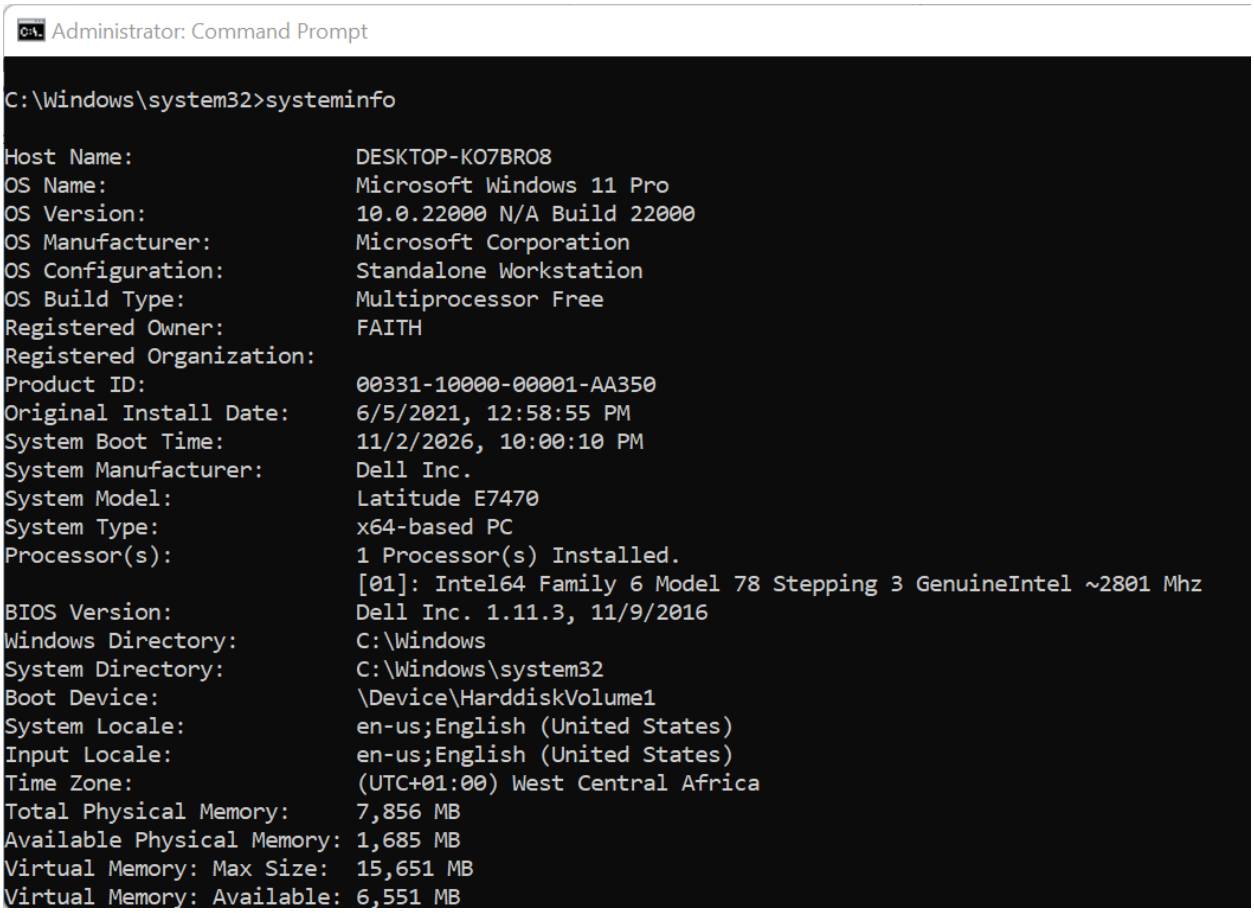


Figure 3- System info using cmd

A (ii)- Capture storage information

- Open Command Prompt or PowerShell.
- Run one of the command sets below. Capture screenshots of storage model, size, partitions, and file systems.

wmic diskdrive get model,size,serialnumber
wmic logicaldisk get caption,filesystem,size,freespace,volumename

PowerShell alternative:
Get-Disk
Get-Partition
Get-Volume

```
C:\Windows\system32>wmic diskdrive get model,size,serialnumber
Model                          SerialNumber          Size
LITEON CV3-8D256-11 SATA 256GB  TW0FVPC4LOH0073U04R1  256052966400


C:\Windows\system32>wmic logicaldisk get caption,filesystem,size,freespace,volumename
Caption  FileSystem  FreeSpace    Size          VolumeName
C:       NTFS        13024911360  255474003968
```

Figure 4- My Windows Storage Information

A (iii)- Map evidence sources to system layers

- Create a table in your report showing at least five system layers/sources and the type of evidence that can be generated from each one.
- Use this guide: User activities = Applications + Operating System + Disk/Memory.

Layer / Source	Example Evidence	Where It May Be Found	Why It Matters
User layer	Opened files, typed usernames	Recent files, browser history,	Links user action to system activity
Application layer	Local storage caches, session tokens, Office document metadata, browser cache, chat logs	Application folders and databases	It ties activity to intent- shows user web activity, what software created or used the data, what was searched, downloaded, sent or opened

Layer / Source	Example Evidence	Where It May Be Found	Why It Matters
OS layer	Event logs, Windows registry hives, installed programs, user profiles	Windows logs, registry, Linux /var/log	Shows system events – you can map program execution history, user account activities, system startup configurations, and anomalous administrative actions
Network layer	DNS cache entries, ARP tables, active TCP/UDP connections, DHCP lease histories, and proxy/firewall transaction logs	C:\Windows\System32\drivers\etc\, registry network keys, router logs	Traces lateral movement, Command and Control (C2) infrastructure communication, data exfiltration channels, and external IP interactions.
Memory/process layer	Running processes, open network connections, injected code, command lines, encryption keys, passwords, clipboard, fileless malware, PIDs.	Live memory capture, pagefile.sys, crash dumps, process list	Shows live system behavior and temporary artifacts that can disappear upon reboot. It also exposes malware that never touches disk.

- Explain why evidence may exist in more than one layer.

Typically in real life situations, evidence barely exist in isolation. This is because operating systems, applications, and hardware interact continuously, and evidence of a single event most times leaves distinct footprints across multiple layers. For example, if a malicious executable is downloaded from the internet, the malicious .exe can exist on disk within a user's directory or temp folder. Once executed, the process runs in RAM, potentially injecting code into legitimate processes (like explorer.exe or svchost.exe), holding active network sockets, or keeping decrypted strings in memory. Furthermore, Windows Event Logs will also show records of the parent-child process creation hierarchy, and Registry keys may reflect execution persistence. It is in bringing all evidence together that forensic analysts get a bigger picture of what happened.

Task B – Program, Process and Thread Investigation

B (i). Observe running processes

- Open Task Manager on Windows, or use Command Prompt/PowerShell.
- Capture a screenshot showing running processes.

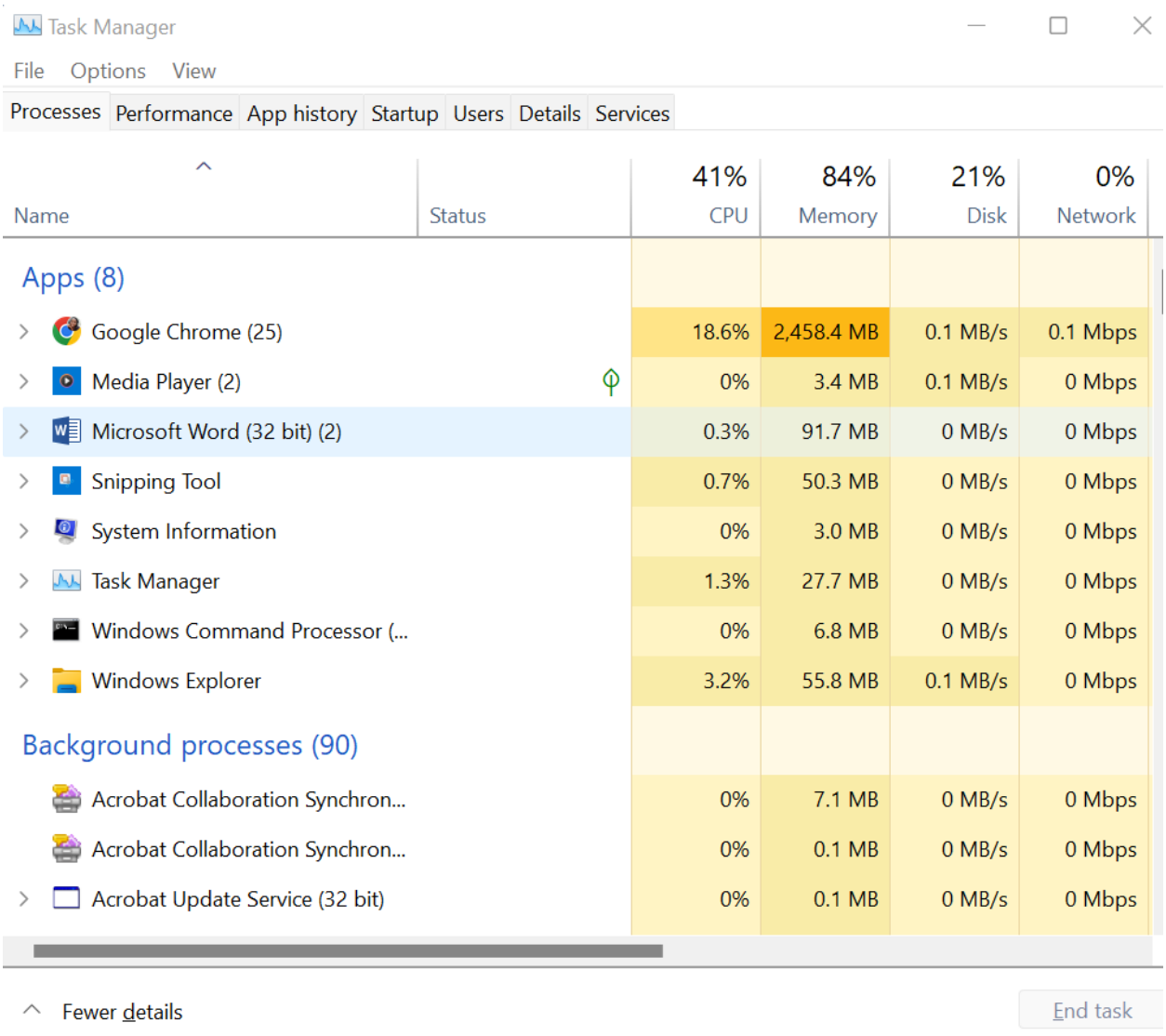


Figure 5- Windows taskbar

- Select three processes and record their process names, PID values, and likely purpose.

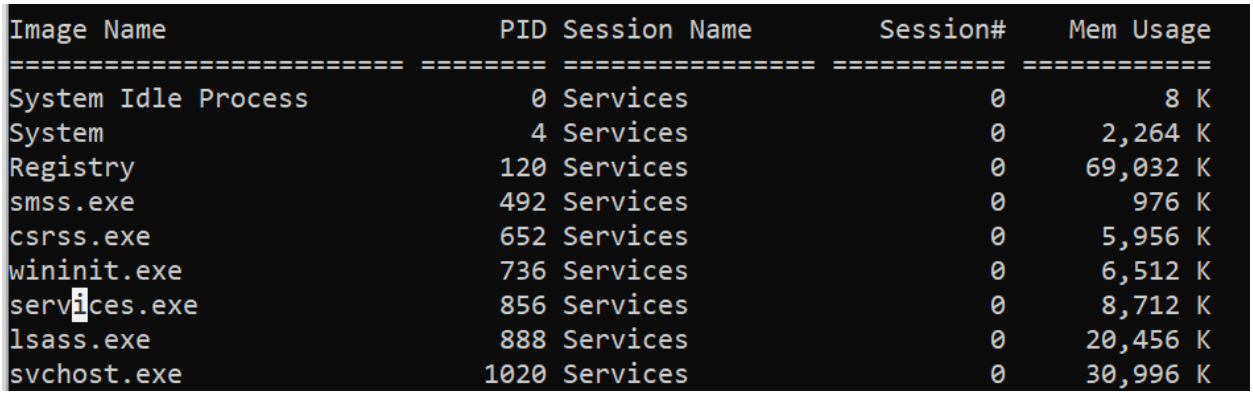


Image Name	PID	Session Name	Session#	Mem Usage
System Idle Process	0	Services	0	8 K
System	4	Services	0	2,264 K
Registry	120	Services	0	69,032 K
smss.exe	492	Services	0	976 K
csrss.exe	652	Services	0	5,956 K
wininit.exe	736	Services	0	6,512 K
services.exe	856	Services	0	8,712 K
lsass.exe	888	Services	0	20,456 K
svchost.exe	1020	Services	0	30,996 K

Figure 6- Processes running in my computer

Process	PID	Purpose
Registry	120	It is responsible for handling all registry hive data operations, caching registry keys and values in memory, and servicing read/write requests from both user-mode applications and the operating system kernel.
Ssms	492	The session manager subsystem is responsible for initializing the user session environment. This includes setting up system environment variables, starting the kernel and user modes of the Win32 subsystem (csrss.exe), and launching subsequent session initialization tasks like winlogon.exe.
wininit	736	The windows initialization process (wininit) acts as a container and manager for critical background services that must run throughout the lifetime of the OS. It initializes the Window Station (WinSta0), starts the Service Control Manager (services.exe), the Local Security Authority Subsystem Service (lsass.exe), and the Local Session Manager (lsm.exe).

B (ii). Explain the difference

- Explain the difference between a program, a process, and a thread in your own words.

A program is a set of codes or instruction in a system that is intended to fulfil a certain task. It sits idly or passively in the disk until executed. A process is a program that has been loaded into memory along with all the resources it needs to operate. It is the running instance of a program that has been loaded into memory (RAM) and is managed by the operating system. A thread is the smallest unit of execution within a process. A process may have a single or multiple threads running concurrently.

- **Explain why process information may be useful during malware analysis or live forensic triage.**

Process information is one of the most critical information during malware analysis and live forensic triage. This is because a program must become a process to actually execute instructions on a system, and monitoring active processes provides an immediate window into malicious activity.

Process information is important for several reasons: it can be used to identify execution and persistence like tracing parent-child relationship, locating binary source, etc. We use processes to also spot malware renamed as legitimate processes running in a different environment. Process metadata can also reveal the privilege from which a code may be running.

Task C - Disk Sector and Disk Size Calculation

C (i)- **Reproduce the slide calculation**

Sectors per track = 400, heads = 12, cylinders = 17000, and bytes per sector = 512.

Item	Formula / Value	Answer
Total sectors	400 x 12 x 17000	81,600,000 sectors
Total bytes	81,600,000 x 512	41,779,200,000 bytes
Kilobytes	41,779,200,000 / 1024	40,800,000 KB
Megabytes	40,800,000 / 1024	39,843.75 MB
Gigabytes	39,843.75 / 1024	38.91 GB

C (ii) - **Calculate your own disk or partition size**

- Use your partition table or disk details to identify at least two partitions.

```
Partition ###  Type              Size     Offset
-------------  ----------------  -------  -------
Partition 1    Primary             50 MB  1024 KB
Partition 2    Primary            237 GB    51 MB
Partition 3    Recovery           506 MB   237 GB
```

Figure 7- My computer partitions

- For each partition, record the start sector, number of sectors, sector size, and file system where available.
- Calculate the size of each partition in GB using the formula above.

Partition 1:

- Sector size- 512 bytes
- Start sector- Offset/ sector size = 1024/512 = 2
- Number of sectors- Size/ sector size = 50MB (52,428,800 bytes) = 52,428,800/512 = 102,400
- Partition size (GB) – number of sector * 512 /1024/1024/1024

102,400*512/1024/1024/1024 = 0.048828125 ≈ 0.05 GB

Item	Formula / Value	Answer
Sector size		512 bytes
Start sector	1024/512	2
Total sector	52,428,800/ 512	102,400 sectors
Total bytes	102,400 * 512	52,428,800 bytes
Kilobytes	52,428,800/ 1024	51,200 KB
Megabytes	51,200/ 1024	50 MB
Gigabytes	50/1024	0.048828125 ≈ 0.05 GB

Partition 2:

- Sector size- 512 bytes
- Start sector- Offset/ sector size = 51 MB (52,224 bytes) 52,224/512 = 102
- Number of sectors- Size/ sector size = 237 GB (254,476,812,288 bytes) = 254,476,812,288/512 = 497,025,024
- Partition size (GB) – number of sector * 512 /1024/1024/1024

497,025,024*512/1024/1024/1024 = 0.048828125 ≈ 237 GB

Item	Formula / Value	Answer
Sector size		512 bytes
Start sector	52,224/512	102
Total sector	254,476,812,288 / 512	497,025,024 sectors
Total bytes	497,025,024* 512	254,476,812,288 bytes

Item	Formula / Value	Answer
Sector size		512 bytes
Kilobytes	254,476,812,288 / 1024	248,512,512 KB
Megabytes	248,512,512/ 1024	242,688 MB
Gigabytes	242,688/1024	237 GB

Task D - Safe Partition Table Inspection

D (i)- Windows method - Disk Management, PowerShell and DiskPart

- Open Disk Management by pressing Windows + R, typing diskmgmt.msc, and pressing Enter.
- Capture a screenshot showing your disk layout and visible partitions.

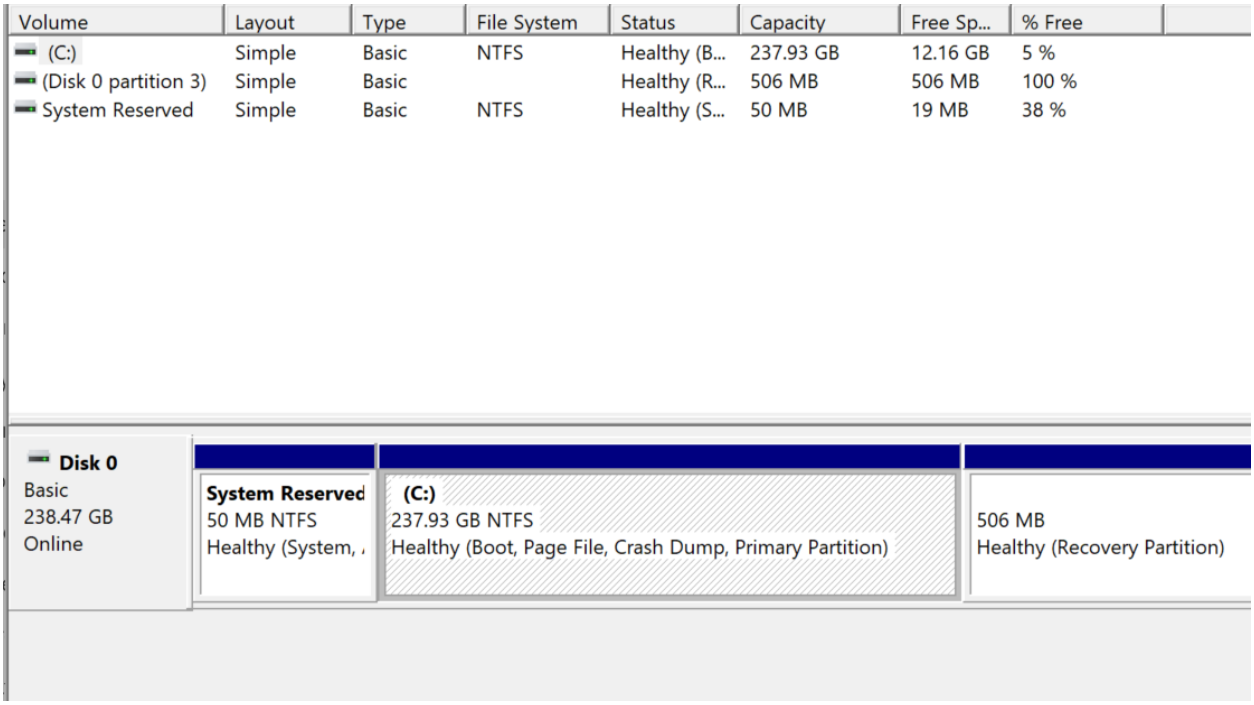


Figure 8- My computer's disk partition on Disk Management

- Open PowerShell as normal user or administrator if required and run the safe read-only commands below.
- Open Command Prompt as administrator and run DiskPart only for listing disks and partitions. Do not run create, delete, clean, format, active, or convert commands.

PowerShell:
Get-Disk
Get-Partition
Get-Volume

DiskPart safe listing only:
diskpart
list disk
select disk 0
list partition
exit

```
PS C:\Windows\system32> Get-Disk

Number Friendly Name Serial Number                  HealthStatus         OperationalStatus      Total Size Partition
                                                                                                           Style
------ ------------- -------------                  ------------         -----------------      ---------- ----------
0      LITEON CV3... TW0FVPC4LOH0073U04R1           Healthy              Online                  238.47 GB MBR


PS C:\Windows\system32> get-partition


   DiskPath: \\?\scsi#disk&ven_liteon&prod_cv3-8d256-11#4&2b6473a9&0&000200#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

PartitionNumber  DriveLetter Offset                                        Size Type
---------------  ----------- ------                                        ---- ----
1                            1048576                                     50 MB IFS
2                C           53477376                                   237.93 GB IFS
3                            255527485440                                506 MB Unknown


PS C:\Windows\system32> get-volume

DriveLetter FriendlyName    FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining      Size
----------- ------------    -------------- --------- ------------ ----------------- -------------      ----
            System Reserved NTFS           Fixed     Healthy      OK                     19.34 MB     50 MB
C                           NTFS           Fixed     Healthy      OK                     11.59 GB 237.93 GB
                            NTFS           Fixed     Healthy      OK                     85.14 MB    506 MB
```

Figure 9- My computer's disk partition using PowerShell

```
PS C:\Windows\system32> diskpart

Microsoft DiskPart version 10.0.22000.653

Copyright (C) Microsoft Corporation.
On computer: DESKTOP-KO7BRO8

DISKPART> list disk

  Disk ###  Status         Size     Free     Dyn  Gpt
  --------  -------------  -------  -------  ---  ---
  Disk 0    Online          238 GB  1024 KB

DISKPART> select disk 0

Disk 0 is now the selected disk.

DISKPART> list partition

  Partition ###  Type              Size     Offset
  -------------  ----------------  -------  -------
  Partition 1    Primary             50 MB  1024 KB
  Partition 2    Primary            237 GB    51 MB
  Partition 3    Recovery           506 MB   237 GB

DISKPART> exit
```

Figure 10 - My computer's disk partition using PowerShell (diskpart)

- Open Terminal in your Linux/Kali VM or host.
- Run the commands below and capture screenshots showing disk, partition, size, and file system information.

```
lsblk -o NAME,SIZE,TYPE,FSTYPE,MOUNTPOINT
sudo fdisk -l
sudo parted -l
blkid
df -h
```

```
┌──(kali㉿kali)-[~]
└─$ df -Th
Filesystem     Type      Size  Used Avail Use% Mounted on
udev           devtmpfs  924M     0  924M   0% /dev
tmpfs          tmpfs     194M  1.2M  192M   1% /run
/dev/sda1      ext4       79G   17G   58G  23% /
tmpfs          tmpfs     966M  4.0K  966M   1% /dev/shm
none           tmpfs     1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs          tmpfs     966M  124K  966M   1% /tmp
none           tmpfs     1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs          tmpfs     194M  108K  194M   1% /run/user/1000
```

Figure 11- File system types on my Linux Machine

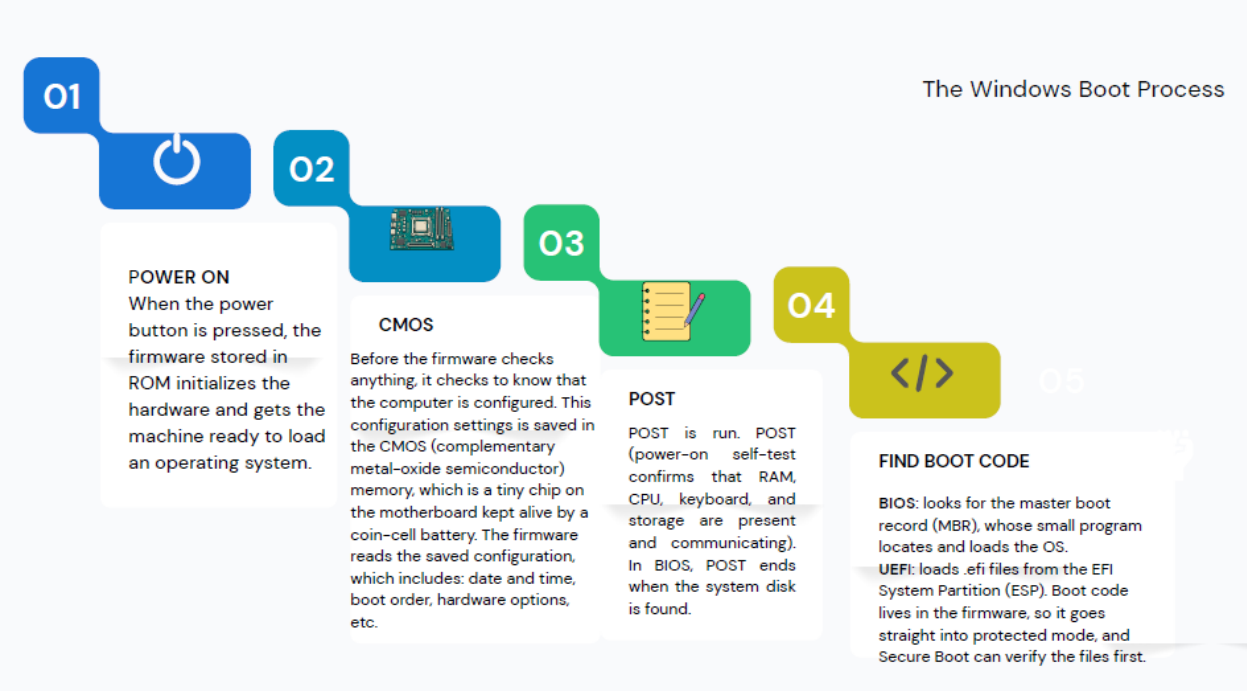
Task E - MBR, GPT and Boot Process Explanation

E (i)- Explain the boot process

- Create a simple flow diagram or numbered list explaining how a PC boots.
- Your explanation must include power-on, POST, BIOS/UEFI, CMOS/firmware settings, MBR/GPT or EFI boot records, boot loader/manager, kernel loading, and operating system control.

A high-level explanation of how a computer boots is this: power is sent to the hardware, firmware wakes up, the hardware gets checked, and the operating system is loaded into memory.

This is a graphical explanation:





- State whether your own system uses BIOS/Legacy or UEFI based on system information evidence.

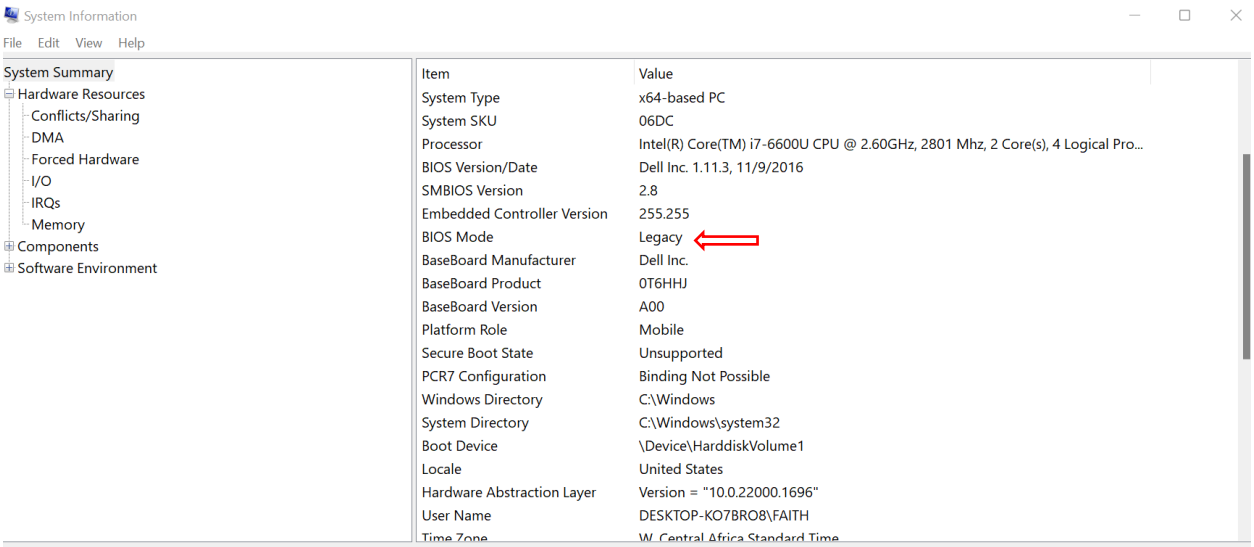


Figure 11- BIOS system information for my computer

Task F - File System Identification and Interpretation

- Identify the file system used by at least two volumes or partitions on your system.
- For Windows systems, identify whether volumes use NTFS, FAT/FAT32, exFAT, or another file system.

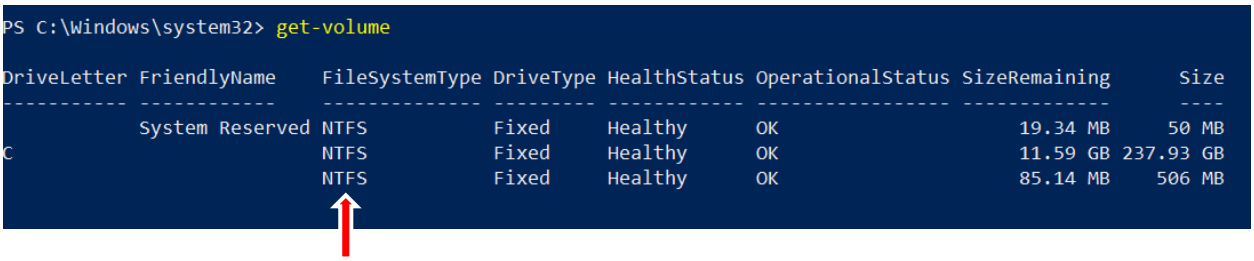


Figure 11- Filesystem of my computer's partitions

The partitions in my Windows system all use the NTFS file system type.

Reflection:

Understanding how a PC boots, how disks are partitioned, and how processes and threads run lets forensic analyst know what normal looks like, so that tampered data, hidden data, and malware stand out. It also guides correct evidence handling, such as imaging a drive without booting it, and helps recover deleted files and rebuild reliable timelines.